

SYED AFFAN ALI

Mobile Application Developer • IoT & Connected Systems • Embedded Systems & Robotics

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PROFESSIONAL SUMMARY

Computer Systems Engineering graduate with 1.5+ years of professional mobile application development experience, specializing in Flutter/Dart, responsive UI, API integration, and offline-first applications. Strong foundation in embedded systems, IoT, sensors, and robotics, with experience building projects that connect hardware, data, and software. Interested in developing practical mobile applications for connected devices and intelligent systems while continuing to build skills in AI/ML and computer vision.

WORK EXPERIENCE

Mobile Application Developer

Gentec Pvt. Ltd. | Karachi, Pakistan | 11/2024 – Present

- Contributing to production-level in-house mobile application development, collaborating within cross-functional agile teams following industry-standard version control and CI/CD workflows.
- Implementing feature modules using Flutter/Dart with emphasis on clean architecture, scalable state management, and code reusability across platforms.
- Performing structured task delivery within sprint cycles under senior developer mentorship, consistently meeting deadlines in a fast-paced development environment.
- Participating in daily standups, code reviews, and retrospectives — strengthening collaborative engineering practices.

Mobile Application Developer – IoT & Connected Applications

Wamtsol Pvt. Ltd. | Karachi, Pakistan | 02/2024 – 10/2024

- Developed and maintained cross-platform Flutter applications, delivering responsive, pixel-accurate interfaces and reusable components for production features.
- Implemented Provider/Bloc state management and structured application architecture to support data-driven features and scalable mobile workflows.
- Integrated device capabilities including GeoLocation, Camera, and Media Access, building mobile features that interact with device hardware and field data.
- Used IsarDB and SharedPreferences for local data storage and Dart isolates for background processing and synchronization, supporting reliable operation in connected and low-connectivity scenarios.
- Worked with data-oriented mobile workflows relevant to connected-device and IoT use cases, including remote data capture, monitoring, and synchronization.

Flutter Front-End Developer – Connected & IoT Applications

BaijiSoft Technologies | Karachi, Pakistan | 08/2023 – 01/2024

- Translated Figma UI/UX designs into high-fidelity Flutter interfaces for data-driven mobile applications, focusing on reusable components and smooth user interaction.
- Implemented REST API communication with GET, POST, and multipart requests, supporting the exchange of application and device-related data between mobile interfaces and backend services.
- Built clear data presentation and interaction flows suitable for monitoring, status, and control-oriented applications, complementing a broader background in IoT and embedded systems.
- Ensured consistent Flutter UI behavior across Android/iOS and different screen sizes while keeping interfaces practical for field and connected-system use cases.

Mobile Application Developer Intern

Zensoft Pvt. Ltd. | Karachi, Pakistan | 06/2023 – 07/2023

- Built a production Sugar Cane Trolley Registration App using Flutter with native GeoLocation and Image Picker APIs, deployed for real field operations.

- Implemented nested navigation, structured state management, and integrated Google Maps SDK for location-aware features.
- Engineered offline-first background data upload pipeline using Dart isolates, ensuring data integrity without active connectivity.
- Applied unit testing, async programming, and app lifecycle management practices — delivering a robust, error-resilient application.

EDUCATION

Bachelor of Engineering — Computer Systems
 Mehran University of Engineering & Technology (MUET) | Jamshoro, Pakistan | 2019 – 2023

TECHNICAL SKILLS

Mobile Development	Flutter · Dart · Provider · Bloc · IsarDB · SharedPreferences · Dart Isolates · REST APIs · JSON · HTTP/Multipart
AI / ML	Python · TensorFlow · PyTorch · Scikit-learn · Hugging Face Transformers · LangChain · RAG Pipelines · Prompt Engineering
Computer Vision	OpenCV · Image Processing · Object Detection · Gesture Recognition · Real-time Video Analysis
Agentic AI	LangChain Agents · Tool Use · LLM Orchestration · OpenAI API · Autonomous Pipelines
Embedded & Robotics	Arduino · C / C++ · Microcontroller Interfacing · Sensor Integration · Gesture-Controlled Robotics · IoT
Languages	Dart · Python · C · C++ · JavaScript (basic)
Tools & Practices	Git · GitHub · Figma · VS Code · Agile/Scrum · Unit Testing · SOLID Principles
Languages (Spoken)	Urdu (Native) · English — IELTS Band 6.5 (Proficient)

PROJECTS

IoT-Based Water Quality Monitoring System | [Arduino](#) · [Python](#) · [Sensors](#) · [IoT](#) · [Data Visualization](#)

- Designed and deployed a real-time water quality monitoring system using Arduino microcontrollers with pH, turbidity, and temperature sensors.
- Built a Python-based data pipeline to collect, normalize, and visualize sensor readings, enabling anomaly detection and threshold-based alerting.
- Implemented wireless data transmission for remote monitoring, demonstrating end-to-end IoT system architecture from hardware to dashboard.

Gesture-Controlled Robotic Arm | [Arduino](#) · [C++](#) · [Servo Motors](#) · [Accelerometer](#) · [Computer Vision](#)

- Engineered a gesture-controlled robotic arm using accelerometer-based hand tracking and servo motor actuation, achieving real-time responsive control.
- Programmed motion interpolation algorithms in C++ for smooth, precise arm movements with configurable sensitivity and range-of-motion limits.
- Explored computer vision-based control as an extension, using OpenCV for hand landmark detection as an alternative input modality.

AI-Powered Document Q&A Agent | [Python](#) · [LangChain](#) · [OpenAI API](#) · [FAISS](#) · [RAG](#) · [Streamlit](#)

- Built an autonomous RAG-based agent that ingests PDF/text documents, creates FAISS vector embeddings, and answers natural language queries with source citations.
- Integrated LangChain's ConversationalRetrievalChain with memory to maintain context across multi-turn dialogues, enabling coherent document exploration.
- Deployed an interactive Streamlit interface for document upload, query input, and highlighted answer rendering — demonstrating full LLM application development lifecycle.

Sugar Cane Trolley Registration App | [Flutter](#) · [Dart](#) · [GeoLocation](#) · [Image Picker](#) · [Dart Isolates](#) · [IsarDB](#)

- Delivered a production field application for real-time trolley registration using native GeoLocation and Image Picker APIs for geo-tagged photo capture.
- Implemented offline-first architecture with IsarDB local storage and background sync via Dart isolates—ensuring reliable data capture in low-connectivity field conditions.
- Deployed to active field operations, demonstrating end-to-end ownership from requirements gathering through production delivery.

AI-Based IoT Energy Monitoring & Anomaly Detection | ESP32 · Python · Machine Learning · IoT · Sensors

Built an IoT prototype to collect electrical measurements from an ESP32 and stream readings for remote monitoring and analysis.

Applied a lightweight anomaly-detection model to identify unusual consumption patterns and presented the results through a simple monitoring interface.

Smart Greenhouse Monitoring & Prediction System | ESP32 · Python · Sensors · ML · IoT

Used temperature, humidity, and light sensors with an ESP32 to collect environmental data and make readings available for remote monitoring.

Applied a simple machine-learning model to predict near-term conditions and generate threshold alerts, connecting sensor data with practical decision support.

RESEARCH INTERESTS

AI-Driven Robotics & Autonomous Systems • Computer Vision for Real-World Applications • Multimodal AI & Human-Robot Interaction • Edge AI & Embedded Machine Learning (TinyML) • Large Language Models & Agentic AI • IoT & Intelligent Sensing Systems

CERTIFICATIONS & TRAINING

Mobile App Development with Flutter | MUET | 07/2022 – 12/2022

Arduino & Robotics | Creativo, Karachi | 04/2021 – 06/2021

IELTS English Proficiency | Band Score: 6.5

RESEARCH OBJECTIVE

To pursue a Master's degree in Artificial Intelligence, Robotics, or Intelligent systems—bridging a strong foundation in mobile application development and embedded hardware with cutting-edge AI research. Committed to contributing to innovative solutions at the intersection of computer vision, autonomous robotics, and edge AI, with the goal of producing impactful, publishable research and building next-generation intelligent systems.